

AUTONOMOUS FACE-TRACKING LASER TURRET

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Implementation

How it was made

- I made my robot as a pan tilt robot where one motor looks left and right and 2 two other motors to look up and down independently from each other
- I connected it to my battery station as it was essentially a stationary machine .
- The Arduino is connected to the breadboard for shared power and connection and then connected to my laptop to input Arduino code

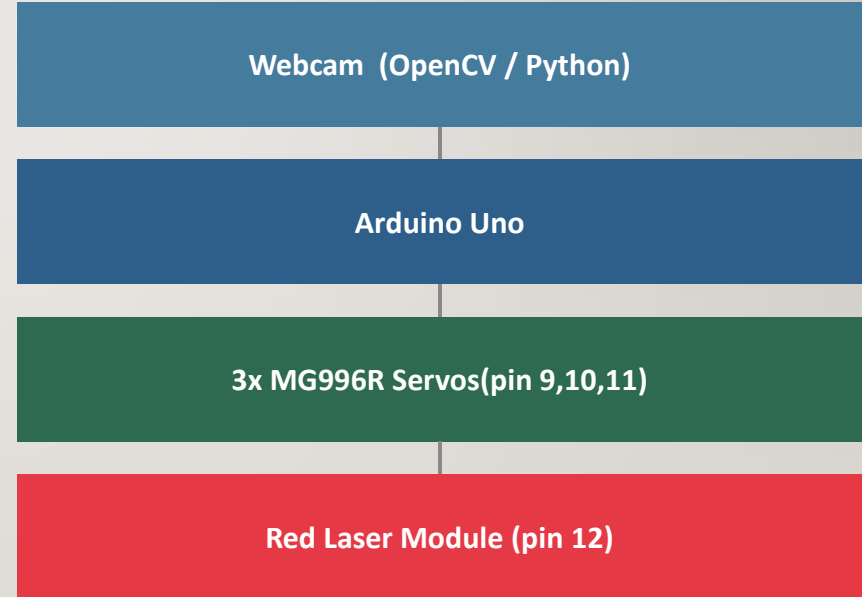


PROJECT OVERVIEW

What It Does

- Webcam feeds live video to Python, where OpenCV detects the face and finds the laser dot
- Python calculates the gap between the face center and laser, then sends correction values to the Arduino over USB serial
- Arduino drives 3 servos via PWM to close that gap, while toggling the laser on pin 12 for detection
- When no face is found, the turret sweeps autonomously until one appears

Hardware Stack



USB Serial / PWM signals / Digital GPIO



HOW IT WORKS

SWEEP MODE

Pan servo across frame

Detects face

Face detected?

NO -- loop back

YES ---->

< lost lock: return to SWEEP

TRACK MODE

Blink laser: OFF then ON

Subtract frames to find the laser

Compute the difference between the face scan and laser scan

Send correction to Arduino to correctly move motors

If lost face

Costs

Overall the total cost was about 120 dollars.

It would have cost more if I did not have the electronics materials to power and connect to the motor



- METERXITY 10-Pack MF83ZZ Flanged Ball Bearing
- DaFuRui 20Pcs Servo Horn Metal Aluminum 25T Silvery Servo Disc
- 3x8x3mm Steel Deep Groove Ball Bearing, Model Bearings MR83ZZ L-830ZZ Pack of 10
- DaFuRui 5 Sets Pan Tilt Servo Mount Bracket for MG995 MG996R S3003 U-Shaped L-Shaped Steering Gear Bracket Robot Car Boat
- Aideepen [6-Pack] MG996R Metal Gear High Speed Torque Digital Servo Motor for JR RC Helicopter Car Boat Robot
- [100pcs M3 x 6mm 12.9 Grade Alloy Steel Hex Socket Head Cap Screws Bolts, Black Oxide Finish](#)
- HiLetgo 5pcs DC 5V Laser Transmitter Module Wave Length 650 nm for Arduino PIC AVR
- Aideepen [6-Pack] MG996R Metal Gear High Speed Torque Digital Servo Motor for JR RC Helicopter Car Boat Robot

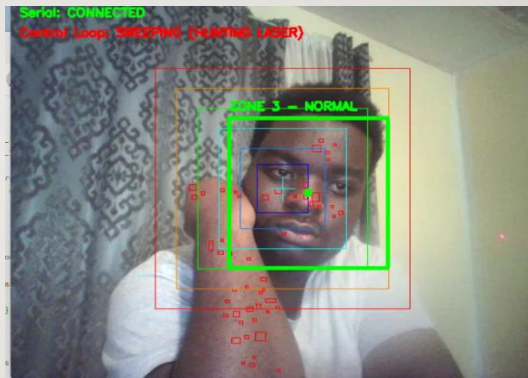
CHALLENGES

(1) Laser Detection on Dark Skin

The Camera misidentified my skin tone as the laser dot
Due to my red undertones, causing false locks instead
of tracking the actual laser.

Red laser reflection on darker skin is too dim for
standard color-based filters to separate reliably.

Black also absorbs lasers so all my hair will absorb the
laser when correcting



(2) No Access to 3D Printer

UConn Library 3D printing lab was closed on the only
available day.

The designed servo base could not be printed -- turret
had no rigid mounting frame.



METHODS & SOLUTIONS

(1) Detection Fix: Rewrite code

My old approach with the coding was HSV color filtering for red pixels. This failed because the laser on my skin was too dim and was fighting my color undertones.

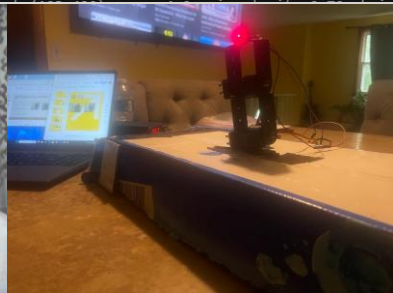
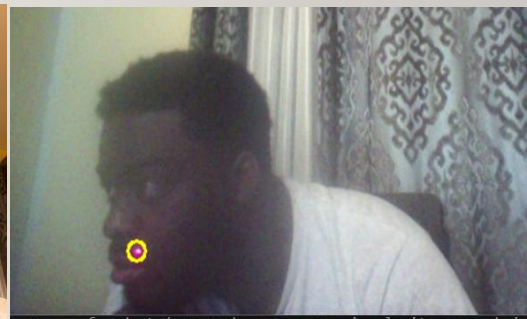
My new approach was to blink the laser OFF and capture a frame, then blink it ON and capture another. Subtract the two frames so the laser stands out regardless of my skin tone or surface.

(2) Physical Fix: Built Cardboard Base

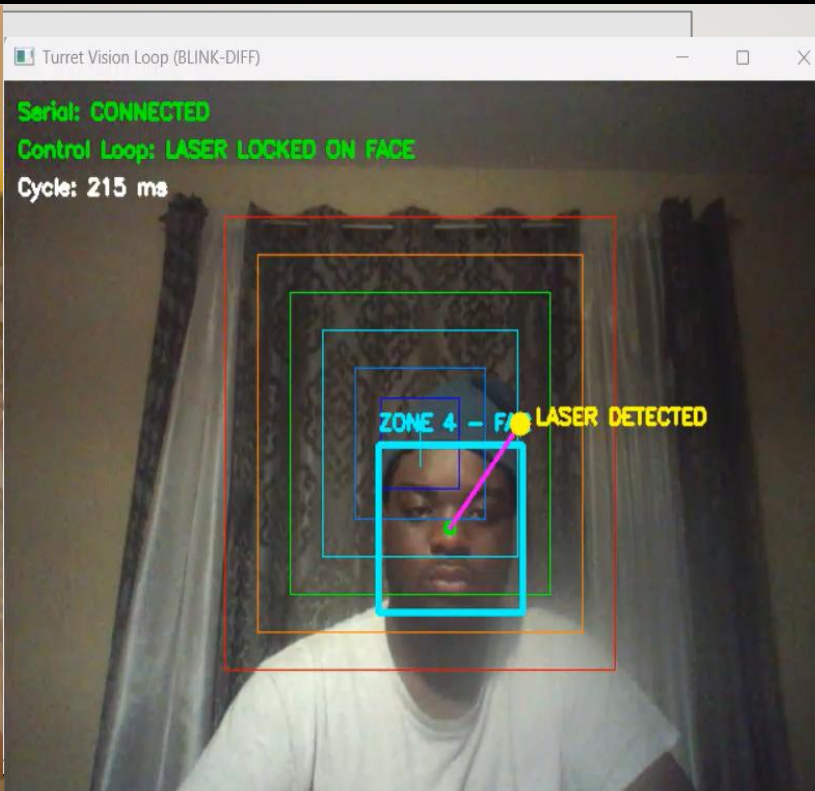
I could not 3D print the servo base as the library was closed. Due to this I made my own base frame from cardboard instead, allowing the full turret to be mounted

(3) Physical Fix: Wear a hat

Since my hair absorbs the laser I wear a Hat to cover it. My beard hair is minimal so it should cause minimal problems



LIVE DEMO



What to show:

Turret sweeps autonomously

Face detected -- tracking starts

Laser corrects toward face center

Head moves -- turret follows

Laser detected on skin

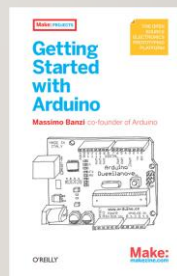
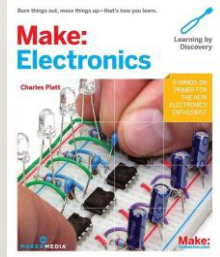
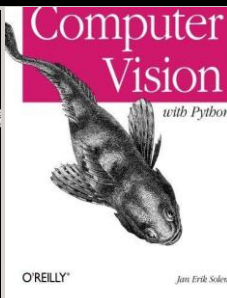
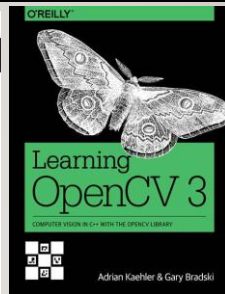
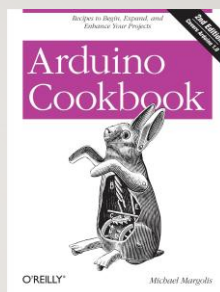
Slight bugs when it comes to my sideburns
beard



Books Used

Here are the books used from O'Reilly Learning library(which is free for UConn Students)

- "Arduino Cookbook" by Michael Margolis** — servo control, serial communication, digital pin output.
- "Learning OpenCV 3" by Bradski & Kaehler** —Contour detection, image thresholding, HSV masking — all in depth.
- "Programming Computer Vision with Python" by Jan Erik Solem** — frame differencing, blob detection, and camera calibration.
- "Make: Electronics" by Charles Platt** — covers how laser modules, transistors, and digital GPIO actually work at the circuit level.
- "Getting Started with Arduino" by Massimo Banzi** — written by Arduino's co-creator. Covers PWM, servo libraries, and serial protocol.



THANK YOU

Questions?

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